CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

Invariant Points and Lines

Questions by topic 9 questions · 75 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

Question 1	*4 marks*	2
Question 2	*5 marks*	3
Question 3	*9 marks*	4
Question 4	*9 marks*	5
Question 5	*13 marks*	6
Question 6	*8 marks*	7
Question 7	*7 marks*	8
Question 8	*7 marks*	9
Question 9	*13 marks*	10

1. The matrix $\mathbf{M}$ represents the transformation T and is given by

$$\mathbf{M} = \begin{pmatrix} p+3 & -2 \\ p+2 & 1-p \end{pmatrix}$$

(a) In the case when $p = -1$ show that the image of the point $(3, 4)$ under T is the point $(-2, 11)$. **[2]**

(b) In the case when $p = 2$ find the gradient of the line of invariant points under T. **[2]**

2. A linear transformation of the plane is represented by

$$\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$$

Determine the invariant line with negative gradient. **[5]**

3. The matrix of a linear transformation T is

$$\mathbf{M} = \begin{pmatrix} p+2 & -5 \\ -2 & p-1 \end{pmatrix}$$

(a) Find the values of p for which T is not invertible. **[4]**

It is now given that $p = 3$, so that

$$\mathbf{M} = \begin{pmatrix} 5 & -5 \\ -2 & 2 \end{pmatrix}$$

(b) Find the equations of the invariant lines, through the origin, of the transformation represented by $\mathbf{M}$. **[5]**

4. The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \frac{1}{5}\begin{pmatrix} -3 & -4 \\ -4 & 3 \end{pmatrix}$$

You are given that $\mathbf{A}$ represents the transformation T, where T is a reflection in a straight line through the origin, O.

(a) Explain why T must have a line of invariant points. State the geometric significance of this line. **[2]**

(b) By considering the line of invariant points for T, determine the equation of the mirror line.
Give your answer in the form $y = mx + c$. **[2]**

(c) The coordinates of the point P are $(4, 7)$.
By considering the image of P under the transformation T, or otherwise, determine the coordinates of the point on the mirror line which is closest to P. **[3]**

(d) The line with equation $2x + by = 6$ is an invariant line for T.
Determine the value of b. **[2]**

5. The matrix $\mathbf{M}$ is given by

$$\mathbf{M} = \begin{pmatrix} 1 & 0 \\ 0 & 6 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

(a) The matrix $\mathbf{M}$ represents a sequence of two geometrical transformations in the x-y plane. Give full details of the two transformations, stating clearly the order in which they are applied. **[4]**

(b) Write $\mathbf{M}^{-1}$ as the product of two matrices, neither of which is $\mathbf{I}$. **[2]**

(c) Find the equations of the invariant lines, through the origin, of the transformation represented by $\mathbf{M}$. **[7]**

6. A linear transformation of the plane is represented by the matrix

$$\mathbf{M} = \begin{pmatrix} 2 & -1 \\ p & 0 \end{pmatrix}$$

where p is a constant.

(a) Show that the line $x = 0$ is not invariant. For a line through the origin with equation $y = mx$, show that it is invariant only if

$$m^2 - 2m + p = 0$$

Hence find the set of values of p for which the transformation has no invariant lines through the origin. **[5]**

(b) Given that the transformation multiplies areas by 3 and reverses orientation, find the invariant lines. **[3]**

7. The matrix

$$\mathbf{B} = \begin{pmatrix} 1 & -4 \\ 2 & -1 \end{pmatrix}$$

represents the linear transformation T.
Any invariant line for T must be parallel to an invariant line through the origin.

(a) A line through the origin has equation $y = mx$, where m is real. Show that, for $m \neq \frac{1}{4}$, its image under T has equation

$$y = \frac{2-m}{1-4m}x$$

State also the image of the line $y = \frac{1}{4}x$ and the image of the vertical line $x = 0$. **[3]**

(b) Hence prove that T has no invariant lines. **[4]**

8.

$$\mathbf{A} = \begin{pmatrix} b & a-3 \\ 2a-1 & -2 \end{pmatrix}$$

where a and b are non-zero constants.
Given that the matrix $\mathbf{A}$ is self-inverse,

(a) determine the value of b and the possible values of a. **[5]**

The matrix $\mathbf{A}$ represents a linear transformation T of the plane.
Using the smaller value of a from part (a),

(b) show that the invariant points of the linear transformation T form a line, stating the equation of this line. **[2]**

9. The 2×2 matrix $\mathbf{A}$ represents a transformation T which has the following properties.

- The image of the point $(1, -1)$ is the point $(2, 0)$.
- An object shape whose area is 4 is transformed to an image shape whose area is 24.
- T has a line of invariant points.

(i) Find a possible matrix for $\mathbf{A}$. **[8]**

The transformation S is represented by the matrix $\mathbf{B}$ where

$$\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}$$

(ii) Find the equation of the line of invariant points of S. **[2]**

(iii) Show that any line of the form $y = 2x + c$ is an invariant line of S. **[3]**